
Technical Task — Senior AI Engineer
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Document code DT-AI-TASK-01	Version 1.1	Date July 2026	Classification For candidates
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1) Problem statement

You are given a document-grounded question-answering assistant (RAG) over a small set of technical documents. A working but weak baseline is sent along with the task: it runs, but the quality of its retrieval and the trustworthiness of its answers are low. Your job is to diagnose the cause of the weakness and improve it.

This task assesses your ability to diagnose an existing system, design retrieval and hallucination control, design an evaluation, deal with imperfect real data, and be transparent about technical decisions.

The suggested time for this task is at most two working days. If you estimate it will take longer, please let us know before proceeding. If time runs short, submit a well-prioritized partial solution with documented trade-offs rather than an unreviewed “complete” one.

Design constraints: this system must work on-premises and offline (no connection to cloud services), with limited computing resources, and over industrial technical documents. Make your technical decisions with these constraints in mind and justify them in the README.

2) What you will do

Diagnose and improve the accompanying baseline over the attached documents, so that the following requirements are satisfied:

Expected deliverables

- **Diagnosis:** run and examine the baseline, and explain in the README why the quality of retrieval and answers is poor.
- **Retrieval improvement:** improve the pipeline so that answers become more relevant and better grounded (for example: better chunking, returning several results with re-ranking, and, if needed, hybrid semantic-plus-keyword retrieval).
- **Hallucination control (abstention):** when the answer does not exist in the documents, the system must explicitly state that it was not found in the documents instead of fabricating an answer. This is a key requirement.
- **Evaluation:** build a small evaluation set — several questions with the expected reference document/answer, and several questions whose answers are not in the documents (to measure abstention). Report and interpret retrieval quality before and after your improvements (for example, the rate of finding the correct document).

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- README (decision log): including run and test instructions, what was wrong and what you changed and why, the trade-offs and how your decisions relate to the constraints above, and an “AI Usage” section as specified under Submission terms.

An LLM answer-generation layer is not required; you may use a small local model or keep this layer simple or simulated. The main focus is retrieval quality, the abstention logic, and the evaluation.

Structure of each document in corpus.jsonl

Field	Type	Example
id	string	DOC-03
title	string	Compressor C-100 — Specifications
text	string	a short technical passage (a few sentences)

Important note: like real industrial documentation, the accompanying corpus is not perfectly clean. Identifying data-quality issues, choosing an explicit policy for each, and documenting both the issue and the policy in the README are part of the task.

3) Scope

- No user interface, authentication, or production deployment is required.
- Python is our preference (the baseline is in Python as well). You may use any local embedding model of your choice; if downloading a model is restricted, using any available alternative model is allowed — state your choice in the README.
- Focus on: diagnosis, retrieval improvement, abstention, evaluation, and the README.

4) Submission terms

- Deliver the code in a Git repository with an incremental, meaningful commit history — not as a single final commit. If creating a repository is not possible, submitting the project as a ZIP file (without virtual-environment folders such as venv) is accepted. Include a single command or short script that re-runs your evaluation and reproduces every metric you report.
- Using AI tools is allowed and is not penalized, provided it is disclosed. The README must contain an AI Usage section stating: which tools were used, for which parts, what you changed after reviewing their output, and at least one concrete mistake made by the tool that you caught and corrected. If no AI tools were used, state that explicitly. Undisclosed AI use, if discovered, disqualifies the submission regardless of its quality.
- Technical defense session: submissions that reach the final stage are reviewed in a mandatory 45–60 minute online session. You will run your code live, explain and defend your decisions, and extend the solution on new documents provided during the session; the live-extension part is done without AI assistance. No offer is made without this session.

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- Reproducibility: we will run your code and re-compute the metrics you report. Reported results that cannot be reproduced from your submission are disqualifying.
 - Final deliverable: the Git repository address along with the README file.

5) Evaluation criteria

The following matter to us in the evaluation:

- Correct diagnosis and a tangible improvement in retrieval quality.
- Hallucination control: correct abstention when the answer is not in the documents.
- Deliberate handling of conflicting information across documents and of documents with nearly duplicate content.
- Quality of the evaluation design and the interpretation of its results.
- Cleanliness and structure of the code.
- Quality of the README and how the decisions relate to the constraints.
- Clarity and depth of the explanation and live extension in the defense session.
- Reproducibility of reported results and honesty of the AI Usage disclosure.

6) What you are given

- The corpus.jsonl file: 16 short technical documents (one document per line, with the keys id, title, and text), sent to you along with this task.
- The baseline_rag.py file: a simple retrieval baseline that runs but produces poor answers, sent to you along with this task.

We wish you every success.